A PROJECT REPORT
ON

CYBERSECURITY AWARENESS GAME

Submitted in partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Computer Science & Engineering

Submitted to:

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Project Guide
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CERTIFICATE

This is to certify that the work embodied in this Project entitled **"CYBERSECURITY AWARENESS GAME"** being submitted by **"Swati Jain (24EARCS171), Shruti Kumari (24EARCS159), Siya Gumber (24EARCS162), Upasna Yadav (24EARCS177), Sneha (24EARCS163)"** in partial fulfillment of the requirement for the award of **"Bachelor of Technology in Computer Science & Engineering"** to Rajasthan Technical University, Kota (Raj.), during the academic year 2025–26, is a record of a bona fide piece of work carried out by them under our supervision and guidance in the **Department of Computer Science & Engineering, Arya College of Engineering & Information Technology, Jaipur.**

PROJECT GUIDE
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DECLARATION

We "**Swati Jain, Siya Gumber, Upasna Yadav, Sneha and Shruti Kumari**", students of Bachelor **of Technology** in **Computer Science & Engineering, session 2026-27. Arya College of Engineering & Information Technology, Jaipur**, here by declare that the work presented in this Project entitled "**Cybersecurity Awarness game**" is the outcome of our own work, is bonafide and correct to the best of our knowledge and this work has been carried out taking care of Engineering Ethics. The work presented does not infringe any patented work and has not been submitted to any other University or anywhere else for the award of any degree or any professional diploma.

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ABSTRACT

The Cybersecurity Awareness Game is an interactive educational application designed to create awareness about cybersecurity and online safety. In today's digital world, people use the internet for education, communication, online transactions, and sharing information. However, the increasing use of digital platforms has also created risks such as phishing attacks, online fraud, password theft, and misuse of personal information. Many users are unaware of these threats and do not know how to protect themselves while using the internet. Therefore, cybersecurity awareness is essential for safe and responsible use of digital technology.

The main objective of this project is to help users understand basic cybersecurity concepts through an interactive game-based learning approach. Traditional learning methods mainly provide theoretical information, which may sometimes be difficult for beginners to understand. This project aims to make learning more interesting by combining cybersecurity information with game-related activities. It encourages users to learn about online threats and understand the importance of maintaining digital security.

The game focuses on important cybersecurity topics such as password security, phishing awareness, protection of personal information, online fraud, and safe internet usage. These topics help users recognize common risks and understand the precautions required to avoid them. For example, users can learn the importance of creating strong passwords, avoiding suspicious links, and not sharing confidential information with unknown sources. Such awareness can help users become more careful while accessing websites, emails, and other online services.

The application is designed to provide a simple and user-friendly learning experience. Users can participate in cybersecurity-related questions and activities

to test their knowledge and improve their understanding. The interactive format encourages active participation and makes it easier to explore cybersecurity concepts. The game also provides an opportunity for users to identify areas where they need to improve their knowledge of online safety.

One of the main advantages of the Cybersecurity Awareness Game is that it presents important security concepts in an easy and engaging manner. It can be useful for students and other users who want to understand the basics of cybersecurity without requiring advanced technical knowledge. By learning through interactive activities, users can develop better awareness of online risks and understand the importance of responsible digital behaviour.

The project also demonstrates how software applications can be used for educational purposes. It provides practical experience in project planning, application design, and software development. The development of the game allows the project team to apply technical knowledge to a real-world problem and understand the importance of cybersecurity in modern society.

In conclusion, the Cybersecurity Awareness Game aims to improve users' understanding of online threats and promote safer internet practices. It provides an interactive way to learn about cybersecurity and encourages users to protect their personal information while using digital platforms. In the future, the project may be enhanced by adding more questions, levels, scoring features, and additional cybersecurity topics to improve the overall learning experience.

CHAPTER 2

Objectives of the Project

The main objectives of the Cybersecurity Awareness Game are as follows:

1. To create cybersecurity awareness:-

The primary objective is to develop an interactive game that helps users understand cybersecurity and the importance of staying safe while using the internet.

2. To identify cyber threats:-

The game aims to educate users about common cyber threats such as phishing attacks, online fraud, suspicious links, and password theft.

3. To promote safe internet usage:-

The project encourages users to follow safe online practices, avoid suspicious websites, and protect their personal information while using digital platforms.

4. To improve password security awareness:-

Another objective is to explain the importance of strong passwords and encourage users to follow better password security practices.

5. To provide interactive learning:-

The game aims to make cybersecurity learning more interesting through interactive questions and activities that help users understand important security concepts.

6. To protect personal information:-

The project helps users understand the importance of keeping sensitive information private and avoiding sharing personal details with unknown or untrusted sources.

7. To encourage responsible digital behaviour:-

The game aims to develop responsible online habits and help users make safer decisions while accessing websites, emails, and other internet services.

8. To enhance technical skills:-

This project helps the team understand the practical application of software development concepts and demonstrates how an interactive game can be used for educational purposes.

CHAPTER 3

Scope of the Project

The scope of the Cybersecurity Awareness Game is defined as follows:

1) Cybersecurity awareness

The game is designed to create awareness about cybersecurity and help users understand the importance of online safety in their daily digital activities.

2) Awareness of multiple cyber threats

It covers important cybersecurity topics, including:

- Phishing attacks
- Password security
- Online fraud
- Protection of personal information
- Safe internet usage

3) Learning through an interactive game

The project uses a game-based approach to make cybersecurity learning more interesting and understandable for users.

4) User-friendly interaction

The game aims to provide a simple interface through which users can participate in activities and learn basic cybersecurity concepts easily.

5) Educational activities

The game can include activities such as:

- Cybersecurity-related questions
- Identifying potential online threats
- Learning safe online practices

6) Future expansion capability

The project can be expanded in the future by adding:

- More cybersecurity topics and questions
- Multiple game levels
- Score tracking and progress monitoring
- Different difficulty levels
- Additional interactive features

7) Educational purpose

The project mainly focuses on spreading cybersecurity awareness, promoting safe internet usage, and demonstrating how game-based learning can help users understand basic security concepts.

CHAPTER 4

The theoretical background

The theoretical background of this project explains the fundamental concepts related to cybersecurity and the technologies used in developing the Cybersecurity Awareness Game. The project is based on concepts of cyber safety, online security, interactive learning, and software development.

1) Cybersecurity

Cybersecurity refers to the practices used to protect computers, networks, applications, and data from unauthorized access, attacks, and damage. It helps users keep their personal information secure while using digital platforms.

Features of Cybersecurity:

- Protection of personal information
- Prevention of unauthorized access
- Detection of cyber threats
- Safe use of digital platforms
- Protection against online attacks

In this project, cybersecurity is the main concept used to educate users about online threats and the importance of maintaining digital safety.

2) Common Cyber Threats

Cyber threats are activities that can harm computer systems, steal information, or misuse online services. Understanding these threats helps users recognize suspicious activities and take suitable precautions.

Common types of cyber threats:

- Phishing attacks
- Online fraud

- Password theft
- Malware attacks
- Identity theft

In this project, these concepts help users learn about common online risks and understand how to avoid unsafe digital activities.

3) Password Security

Password security is an important part of cybersecurity. Strong passwords help protect user accounts and reduce the risk of unauthorized access.

Important practices for password security:

- Use strong and unique passwords
- Avoid easily guessed passwords
- Do not share passwords with others
- Avoid using the same password for every account
- Keep login information confidential

The project aims to improve users' understanding of password safety and encourage responsible account protection.

4) Phishing Awareness

Phishing is a type of cyberattack in which attackers attempt to trick users into sharing sensitive information through fake emails, messages, websites, or links.

Common warning signs of phishing:

- Suspicious links
- Unknown senders
- Unexpected requests for personal information
- Messages creating urgency
- Fake websites or login pages

In this project, phishing awareness helps users understand the importance of checking messages and links before interacting with them.

5) Safe Internet Usage

Safe internet usage means following practices that protect users from online risks while browsing websites, using social media, or accessing digital services.

Important safety practices:

- Avoid clicking unknown links
- Use trusted websites
- Protect personal information
- Keep applications updated
- Report suspicious online activities

The project focuses on encouraging users to adopt safer internet habits and make informed decisions while using digital platforms.

6) Game-Based Learning

Game-based learning is an educational approach that uses games and interactive activities to explain concepts and improve understanding. It can make learning more engaging by encouraging users to participate actively.

Advantages of game-based learning:

- Increases user engagement
- Encourages active participation
- Makes learning more interesting
- Helps users test their knowledge
- Supports learning through practice

In this project, the game-based approach is used to make cybersecurity education more accessible and interesting for users.

7) Interactive Application

An interactive application allows users to communicate with the system through actions such as selecting options, answering questions, or navigating different sections.

Important features of interactive applications:

- User interaction
- Simple navigation
- Organized presentation of information
- Immediate response to user actions
- Improved user experience

In this project, interactive features can help users participate in cybersecurity-related activities and learn basic online safety concepts.

8) Software Development

Software development is the process of planning, designing, implementing, and testing an application to achieve specific objectives.

Main stages of software development:

- Requirement analysis
- System planning
- Application design
- Implementation
- Testing and improvement

This project provides practical experience in applying software development concepts to create an educational application focused on cybersecurity awareness.

CHAPTER 5

Definition of Problem

In today's digital world, cybersecurity has become an important concern for people using the internet. Many users are unaware of common cyber threats such as phishing, online fraud, password theft, and the misuse of personal information. This lack of awareness can lead to security risks and unsafe online activities.

One of the major problems is the lack of knowledge about cyber threats. Users may not be able to identify suspicious links, fake websites, or fraudulent messages. As a result, they may unknowingly share personal information with unauthorized individuals.

Another problem is weak password security. Many users choose simple passwords, reuse the same password on different platforms, or share their login details with others. These practices increase the risk of unauthorized access to their accounts.

The lack of awareness about safe internet usage is also a major issue. Users may click on unknown links, download suspicious files, or visit untrusted websites without understanding the possible consequences. This can expose their personal information to cybercriminals.

Traditional methods of learning about cybersecurity may not always be engaging or interesting for users. Reading only theoretical information may make it difficult for beginners to understand cyber threats and remember the precautions required to stay safe online.

Additionally, users need a simple and interactive way to test their cybersecurity knowledge. Without practical activities, it may be difficult to understand how to recognize threats and make safe decisions while using digital platforms.

Due to these limitations, there is a need for a system that can:

- Create awareness about cybersecurity and online safety

- Help users understand common cyber threats
- Promote strong password practices
- Encourage safe internet usage
- Provide an interactive and engaging learning experience
- Help users understand the importance of protecting personal information

To overcome these problems, the proposed **Cybersecurity Awareness Game** is designed to improve users' understanding of cybersecurity through a game-based learning approach. It aims to present important cyber safety concepts in a simple and interesting manner.

The proposed game provides an educational approach to learning about online risks and safe digital practices. It aims to increase cybersecurity awareness, encourage responsible internet usage, and help users make safer decisions while using digital platforms.

CHAPTER 6

System Analysis & User Requirements

1) System Analysis

System analysis is the process of studying the existing system and identifying its limitations, requirements, and improvements needed for the new system. It helps in understanding how the system should function to meet user needs effectively.

In the existing learning methods, several issues related to cybersecurity awareness were identified:

- Lack of knowledge about common cyber threats
- Difficulty in identifying phishing messages and suspicious links
- Weak password practices and unsafe online activities
- Limited interactive methods for learning cybersecurity
- Lack of awareness about protecting personal information

To overcome these problems, the proposed Cybersecurity Awareness Game is designed to provide an interactive learning experience. It focuses on helping users understand cyber threats and learn safe online practices.

The system aims to ensure:

- Improved cybersecurity awareness
- Better understanding of online threats
- Increased awareness of password security
- Reduced risks caused by unsafe online practices

- An engaging and user-friendly learning experience

2) Proposed System

The proposed system is a Cybersecurity Awareness Game that helps users learn about cybersecurity through game-based activities. It aims to make learning simple, interactive, and interesting.

The system focuses on:

- Learning about common cyber threats
- Understanding phishing and online fraud
- Improving awareness of password security
- Learning safe internet practices
- Understanding the importance of protecting personal information

3) User Requirements

The system is designed for users who want to improve their knowledge of cybersecurity.

a) User

The user’s requirements include:

- Easy access to cybersecurity learning activities
- Simple and user-friendly interaction
- Understanding common cyber threats
- Learning safe online practices
- Improving awareness of personal information security

4) Functional Requirements

Functional requirements describe the activities the system is expected to perform:

- Provide cybersecurity-related learning content
- Help users understand common online threats
- Present information about safe internet usage
- Support interactive learning activities, according to the implemented features
- Provide a simple interface for users to access the game

5) Non-Functional Requirements

Non-functional requirements define the quality and performance expected from the system:

- The system should be easy to use
- The interface should be simple and understandable
- The system should respond properly to user interactions
- The learning content should be clear and informative
- The system should run efficiently on a compatible device

6) Feasibility Analysis

1. Technical Feasibility

The project is technically feasible because a cybersecurity awareness game can be developed using suitable software development tools and technologies. The final implementation depends on the technologies selected for the project.

2. Economic Feasibility

The project can be developed cost-effectively by using suitable free or open-source development tools and resources, depending on the technologies used.

3. Operational Feasibility

The system is intended to be simple and user-friendly. Users should be able to understand the cybersecurity concepts and interact with the game without requiring advanced technical knowledge.

CHAPTER 7

System Planning

1) Introduction

System planning is an important phase in software development that focuses on organizing resources, defining objectives, and preparing a roadmap for the successful completion of the project. It ensures that the system is developed within the given time, cost, and resource constraints.

In this project, which is a **Cybersecurity Awareness Game**, system planning was carried out to develop an educational game that helps users understand cybersecurity, identify common cyber threats, and learn safe internet practices.

2) Objectives of System Planning

The main objectives of system planning in this project are:

- To clearly define the scope and goals of the system
- To identify the required software and hardware resources
- To ensure proper time management
- To reduce development risks
- To provide a structured approach for development

Proper planning helps in avoiding confusion and ensures the smooth execution of the project.

3) Feasibility Study

Before starting the development, a feasibility study was considered to determine whether the project is practical and achievable.

a) Technical Feasibility

This determines whether the required technology and resources are available.

- Suitable software development tools can be used to develop the game.

- The project can be developed using commonly available computer systems.
- Advanced or costly tools may not be necessary.

Hence, the project is technically feasible, depending on the technologies selected for implementation.

b) Economic Feasibility

This checks whether the project is cost-effective.

- Free or open-source development tools can be used.
- No specialized hardware is necessarily required.
- The project can be developed with limited resources.

Therefore, the project can be economically feasible.

c) Operational Feasibility

This determines whether the system can be easily understood and used by users.

- The game is intended to be simple and user-friendly.
- It helps users learn about cybersecurity threats.
- It promotes awareness of safe internet practices.

Thus, the system is operationally feasible.

4) Resource Planning

Proper resource planning helps ensure the smooth development of the project.

a) Hardware Requirements

- Computer or laptop
- Minimum 4 GB RAM recommended
- Basic storage space

b) Software Requirements

- Suitable programming language and development tools
- Code editor or Integrated Development Environment (IDE)

- Compatible web browser or runtime environment, depending on the implementation

5) Time Planning

Time planning helps in completing the project within the deadline.

The project was divided into different phases:

- Requirement Analysis
- System Design
- Coding and Implementation
- Testing
- Final Report Preparation and Submission

Each phase was assigned time to support organized development and quality work.

6) Risk Management

During planning, possible risks were identified and considered:

- **Coding errors:** identified and corrected through testing.
- **Design issues:** reduced by reviewing the game interface and functionality.
- **Incorrect learning content:** minimized by reviewing cybersecurity information for clarity and accuracy.
- **Compatibility issues:** addressed by checking the game on the intended device or platform.

Planning in advance helps reduce these risks and supports smoother project development.

7) System Schedule

A simple schedule was planned for the different phases of the project.

Phase	Work Done
Phase 1	Requirement Analysis
Phase 2	Game Design and Planning

Phase 3 Coding and Implementation

Phase 4 Testing and Improvements

Phase 5 Final Report Preparation and Submission

8) Summary

System planning plays a vital role in the successful development of the project. It provides a clear direction and helps in managing resources, time, and risks effectively.

In this project, system planning supports:

- Organized development of the Cybersecurity Awareness Game
- Efficient use of available resources
- Identification and reduction of possible errors
- Better organization of project activities

Thus, system planning contributes significantly to the development of the **Cybersecurity Awareness Game**.

CHAPTER 8

Methodologies Adopted and Software Used

1) Introduction

The successful development of any software system depends largely on the methodologies and tools used during its development process. In this project, which is a **Cybersecurity Awareness Game**, suitable methodologies and software tools are considered to develop an interactive, reliable, and user-friendly learning application.

The system is designed to create awareness about cybersecurity topics such as phishing, online fraud, password security, and safe internet usage. Special focus is given to organized development, interactive learning, clear presentation of information, and testing of the application.

2) Methodologies Adopted

a) Modular Approach

The modular approach involves dividing the system into smaller, manageable units called modules. Each module focuses on a particular function or topic.

For example, the game may include sections related to:

- Cybersecurity awareness
- Phishing awareness
- Password security
- Online safety
- Cyber threat identification

Each module focuses on a specific cybersecurity concept, making the application easier to understand and manage.

Advantages of Modular Approach:

- Simplifies the development process

- Makes testing and debugging easier
- Improves code organization
- Supports future improvements and expansion

b) Interactive Learning Approach

The project follows an interactive learning approach to make cybersecurity education more interesting and understandable. Instead of depending only on theoretical information, the game aims to involve users in learning activities.

This approach helps users:

- Understand common cyber threats
- Learn safe online practices
- Improve cybersecurity awareness
- Participate actively in the learning process

Interactive learning can make cybersecurity concepts easier to understand, especially for beginners.

c) Game-Based Learning Approach

Game-based learning uses game elements to support the learning process. In this project, cybersecurity concepts are presented through an educational game format.

The approach aims to:

- Make learning more engaging
- Encourage user participation
- Improve understanding of cybersecurity concepts
- Help users remember important safety practices

The specific game activities depend on the features implemented in the application.

d) User-Friendly Design Approach

A user-friendly design focuses on making the application simple and easy to use.

In this project, the design aims to provide:

- Clear and understandable instructions
- Simple navigation
- Organized cybersecurity information
- Easy interaction with the game

This approach helps users focus on learning rather than dealing with a complicated interface.

e) Waterfall Development Model

The Waterfall Model is a linear and sequential software development process in which development activities are organized into different phases.

Phases of the Waterfall Model:

1. Requirement Analysis

In this phase, the project requirements were identified, including the need to create cybersecurity awareness and explain common online threats.

2. System Design

The structure and flow of the game were planned. This includes organizing cybersecurity topics, designing the interface, and deciding how users will interact with the application.

3. Implementation

In this phase, the application is developed using the selected programming languages and software tools. The planned features and learning content are incorporated into the game.

4. Testing

The application is tested to check whether its features work correctly, the content is understandable, and users can interact with the game properly. Identified errors are corrected.

5. Deployment

The application is prepared to run in its intended environment so that users can access and use the game.

6. Maintenance

Future updates can be made to improve the application, correct errors, and add new cybersecurity topics or learning activities.

3) Software Used

The software tools used in the project depend on the technologies selected for implementation. The following descriptions should be finalized according to the actual tools used to develop the game.

a) Programming Language

A suitable programming language is used to implement the game logic, handle user interactions, and provide the required functionality. The selected language depends on the project's actual implementation.

b) Code Editor or IDE

A code editor or Integrated Development Environment (IDE) is used to write, edit, and manage the source code. It helps organize the project files and makes development easier.

Examples include Visual Studio Code and other compatible development environments, depending on the tools actually used.

c) Web Browser or Runtime Environment

A compatible web browser or runtime environment is used to run and test the application, depending on how the game is developed. It helps verify the interface and user interactions.

d) Testing Tools

Testing is performed to check the working of the game, identify errors, and ensure that the learning content and interactive features behave as expected. The tools used depend on the project implementation.

e) Operating System

The operating system provides the environment for developing and testing the project. The actual operating system used during development should be mentioned in the final report.

4) Summary

In this project, suitable development methodologies and software tools are used to support the development of the Cybersecurity Awareness Game.

The methodologies include:

- Modular approach
- Interactive learning approach
- Game-based learning approach
- User-friendly design approach
- Waterfall development model

These approaches help organize the development process and support the creation of an educational application that promotes cybersecurity awareness.

Thus, the selected methodologies and tools contribute to the development of a structured, understandable, and useful Cybersecurity Awareness Game.

CHAPTER-9

Process and Algorithm

9.1 Introduction

This chapter explains the working process of the system along with the algorithms used in the project. The project is a **Cybersecurity Awareness Game** designed to improve users' knowledge of cybersecurity and online safety.

The system focuses on learning about common cyber threats and understanding safe online practices. The main activities of the system include:

- Learning about cybersecurity
- Identifying common cyber threats
- Understanding password security
- Learning safe internet practices

9.2 Process Involved in the System

The system follows a simple step-by-step process:

Step 1: Start the Application

- The user opens the Cybersecurity Awareness Game.
- The main interface is displayed.

Step 2: Access the Game

- The user accesses the available game or learning activities.
- The instructions are presented to help the user understand the game.

Step 3: Select a Topic or Activity

The user selects an available cybersecurity topic or activity, such as:

- Phishing Awareness
- Password Security
- Online Safety
- Cyber Threat Identification

Step 4: Participate in the Activity

- The user reads the question or situation presented in the game.
- The user selects an answer or performs the available activity.
- The system processes the user's interaction according to the implemented game logic.

Step 5: Display the Result

- The system displays the appropriate response or feedback, if supported by the game.
- The user learns about the cybersecurity concept related to the activity.

Step 6: Continue Learning

- The user can continue with other available activities.
- The user gains knowledge about different cybersecurity topics.

Step 7: Exit

- The user exits the game after completing the activities.

9.3 Algorithm

Algorithm for Starting the Game

I. Start

II. Open the Cybersecurity Awareness Game.

III. Display the main interface.

IV. Allow the user to access the available activities.

V. End.

Algorithm for Selecting an Activity

I. Start.

II. Display the available cybersecurity topics or activities.

III. Allow the user to select an activity.

IV. Display the selected activity.

V. Allow the user to participate in the activity.

VI. End.

Algorithm for Answering a Question

I. Start.

II. Display a cybersecurity-related question, if the game includes questions.

III. Accept the user's answer.

IV. Process the answer according to the game logic.

V. Display the appropriate feedback, if available.

VI. End.

Algorithm for Learning Cybersecurity Practices

I. Start.

II. Display information about a cybersecurity topic.

III. Explain the related threat or safety practice.

IV. Allow the user to review the information.

V. Continue to another topic or exit the game.

VI. End.

9.4 Summary

The Cybersecurity Awareness Game follows a structured process that allows users to access learning activities and understand important cybersecurity concepts. The algorithms describe the basic flow of the application, from starting the game to interacting with its activities.

The system aims to make cybersecurity education simple and engaging while encouraging users to follow safe online practices.

Chapter-10

Source Code

```
package com.cyberaware.backend;

import java.util.List;

public final class AwarenessContent {
    private AwarenessContent() {
    }

    public static List<AwarenessTopic> allTopics() {
        return List.of(
                new AwarenessTopic("Phishing and scams",
   private static AwarenessTopic topic(String title, String advice, String example, String
checklist) {
        return new AwarenessTopic(title, advice, example, checklist);
    }
}

package com.cyberaware.backend;

import com.cyberaware.frontend.CyberAwareUI;
import com.cyberaware.frontend.UiTheme;
import java.sql.SQLException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.swing.JOptionPane;
import javax.swing.SwingUtilities;
import javax.swing.UIManager;
```

```
public final class CyberAwareApp {
    private static final Logger LOGGER =
Logger.getLogger(CyberAwareApp.class.getName());

    private final PlayerDatabase playerDatabase;
    private String playerName = "";
    private int totalPoints;
    private StudyLanguage studyLanguage = StudyLanguage.ENGLISH;
    private long playerId;

    public CyberAwareApp() throws SQLException {
        playerDatabase = new PlayerDatabase();
        playerDatabase.mostRecentPlayer().ifPresent(profile -> {
            playerName = profile.name();
            totalPoints = profile.totalPoints();
            studyLanguage = profile.language();
        });
    }

    public boolean startSession(String name) throws SQLException {
        if (name == null || name.isBlank()) {
            return false;
        }
        PlayerProfile profile = playerDatabase.openPlayer(name, studyLanguage);
        restorePlayer(profile);
        return true;
    }

    public String getPlayerName() {
        return playerName;
    }
```

```
    public void setPlayerName(String playerName) {
        if (playerName == null || playerName.isBlank()) {
            throw new IllegalArgumentException("Player name cannot be blank.");
        }
        this.playerName = playerName.trim();
    }

    public int getTotalPoints() {
        return totalPoints;
    }

    public int getScore() {
        return totalPoints;
    }

    public StudyLanguage getStudyLanguage() {
        return studyLanguage;
    }

    public void setStudyLanguage(StudyLanguage studyLanguage) throws
SQLException {
        if (studyLanguage == null) {
            throw new IllegalArgumentException("Study language cannot be null.");
        }
        if (playerId > 0) {
            playerDatabase.updateLanguage(playerId, studyLanguage);
        }
        this.studyLanguage = studyLanguage;
    }

    public void addPoints(int points) {
```

```
        if (points < 0) {
            throw new IllegalArgumentException("Points cannot be negative.");
        }
        if (playerId <= 0) {
            throw new IllegalStateException("Start a player session before awarding points.");
        }
        try {
            totalPoints = playerDatabase.addPoints(playerId, points);
        } catch (SQLException exception) {
            LOGGER.log(Level.SEVERE, "Could not save the player's score.", exception);
            JOptionPane.showMessageDialog(null,
                    "Your score could not be saved to the local database.\n"
                            + "Please check that the project data folder is writable.\n\n"
                            + exception.getMessage(),
                    "Database save error", JOptionPane.ERROR_MESSAGE);
        }
    }

    private void restorePlayer(PlayerProfile profile) {
        playerId = profile.id();
        playerName = profile.name();
        totalPoints = profile.totalPoints();
        studyLanguage = profile.language();
    }

    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            try {

UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());
```

```
        } catch (ReflectiveOperationException |
javax.swing.UnsupportedLookAndFeelException e) {
            System.err.println("Could not load the system look and feel: " +
e.getMessage());
        }
        UiTheme.install();
        try {
            CyberAwareApp app = new CyberAwareApp();
            new CyberAwareUI(app).setVisible(true);
        } catch (Exception exception) {
            LOGGER.log(Level.SEVERE, "Could not start the application.", exception);
            JOptionPane.showMessageDialog(null,
                    "The application could not be opened:\n\n"
                            + exception.getMessage(),
                    "Application startup error", JOptionPane.ERROR_MESSAGE);
        }
      });
   }
}
```

```
package com.cyberaware.backend;

public final class GameEntity {
    public enum Type {
        MALWARE,
        PHISHING,
        SAFE_PACKET,
        FIREWALL_PULSE,
        SHIELD
    }

    private final Type type;
    private double x;
    private double y;
    private final double speed;
    private boolean active = true;

    public GameEntity(Type type, double x, double y, double speed) {
        this.type = type;
        this.x = x;
        this.y = y;
        this.speed = speed;
    }

    public Type getType() {
        return type;
    }
```

```
public double getX() {
   return x;
}

public double getY() {
   return y;
}

public double getSpeed() {
   return speed;
}

public boolean isActive() {
   return active;
}

public void move(double deltaSeconds) {
   y += speed * deltaSeconds;
}

public void moveUp(double deltaSeconds) {
   y -= speed * deltaSeconds;
}

public void deactivate() {
   active = false;
```

```
    }
}

package com.cyberaware.backend;

import java.nio.file.Files;
import java.nio.file.Path;
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import java.util.Locale;
import java.util.Optional;
import java.util.logging.Level;
import java.util.logging.Logger;

public final class PlayerDatabase {
    private static final Logger LOGGER =
Logger.getLogger(PlayerDatabase.class.getName());
    private final String jdbcUrl;
    private final boolean inMemory;
    private PlayerProfile fallbackProfile;
    private long nextFallbackId = 1L;
```

```
    public PlayerDatabase() throws SQLException {
        this(Path.of("data", "cyberaware.db"));
    }

    public PlayerDatabase(Path databaseFile) throws SQLException {
        boolean sqliteReady = false;
        String url = null;
        try {
            Class.forName("org.sqlite.JDBC");
            Path absoluteFile = databaseFile.toAbsolutePath().normalize();
            Path parent = absoluteFile.getParent();
            if (parent != null) {
                Files.createDirectories(parent);
            }
            url = "jdbc:sqlite:" + absoluteFile;
            try (Connection conn = DriverManager.getConnection(url);
                 Statement stmt = conn.createStatement()) {
                stmt.executeUpdate("""
                        CREATE TABLE IF NOT EXISTS players (
                            id INTEGER PRIMARY KEY AUTOINCREMENT,
                            display_name TEXT NOT NULL,
                            name_key TEXT NOT NULL UNIQUE,
                            total_points INTEGER NOT NULL DEFAULT 0
CHECK(total_points >= 0),
                            language TEXT NOT NULL,
                            last_played TEXT NOT NULL
                        )
```

```
                """);
            }
            sqliteReady = true;
        } catch (Throwable t) {
            LOGGER.log(Level.WARNING, "SQLite JDBC driver not available or failed to initialize: " + t.getMessage()
                    + ". Running with in-memory session.", t);
            sqliteReady = false;
        }
        this.inMemory = !sqliteReady;
        this.jdbcUrl = url;
    }

    public synchronized Optional<PlayerProfile> mostRecentPlayer() throws SQLException {
        if (inMemory) {
            return Optional.ofNullable(fallbackProfile);
        }
        String sql = """
                SELECT id, display_name, total_points, language
                FROM players
                ORDER BY last_played DESC, id DESC
                LIMIT 1
                """;
        try (Connection connection = connect();
             Statement statement = connection.createStatement();
             ResultSet result = statement.executeQuery(sql)) {
```

```
            return result.next() ? Optional.of(readProfile(result)) :
Optional.empty();
        }
    }

    public synchronized PlayerProfile openPlayer(String name,
StudyLanguage language) throws SQLException {
        String normalizedName = name.trim();
        if (inMemory) {
            if (fallbackProfile != null &&
fallbackProfile.name().equalsIgnoreCase(normalizedName)) {
                fallbackProfile = new PlayerProfile(fallbackProfile.id(),
normalizedName, fallbackProfile.totalPoints(), language);
            } else {
                fallbackProfile = new PlayerProfile(nextFallbackId++,
normalizedName, 0, language);
            }
            return fallbackProfile;
        }
        String nameKey = normalizedName.toLowerCase(Locale.ROOT);
        try (Connection connection = connect()) {
            connection.setAutoCommit(false);
            try {
                long playerId = findPlayerId(connection, nameKey);
                if (playerId == 0) {
                    try (PreparedStatement insert = connection.prepareStatement(
```

```
                    "INSERT INTO players(display_name, name_key,
language, last_played) "
                            + "VALUES(?, ?, ?, CURRENT_TIMESTAMP)"))
{
                insert.setString(1, normalizedName);
                insert.setString(2, nameKey);
                insert.setString(3, language.name());
                insert.executeUpdate();
            }
            playerId = findPlayerId(connection, nameKey);
        } else {
            try (PreparedStatement update =
connection.prepareStatement(
                    "UPDATE players SET display_name = ?, language = ?,
"
                            + "last_played = CURRENT_TIMESTAMP
WHERE id = ?")) {
                update.setString(1, normalizedName);
                update.setString(2, language.name());
                update.setLong(3, playerId);
                update.executeUpdate();
            }
        }
        PlayerProfile profile = findPlayer(connection, playerId)
                .orElseThrow(() -> new SQLException("Saved player
could not be reloaded."));
        connection.commit();
```

```
            return profile;
        } catch (SQLException | RuntimeException exception) {
            connection.rollback();
            throw exception;
        }
    }
}

public synchronized void updateLanguage(long playerId, StudyLanguage language) throws SQLException {
    if (inMemory) {
        if (fallbackProfile != null && fallbackProfile.id() == playerId) {
            fallbackProfile = new PlayerProfile(playerId, fallbackProfile.name(), fallbackProfile.totalPoints(), language);
        }
        return;
    }
    try (Connection connection = connect();
         PreparedStatement statement = connection.prepareStatement(
                 "UPDATE players SET language = ? WHERE id = ?")) {
        statement.setString(1, language.name());
        statement.setLong(2, playerId);
        if (statement.executeUpdate() != 1) {
            throw new SQLException("The active player record could not be updated.");
        }
    }
```

```
    }

    public synchronized int addPoints(long playerId, int points) throws SQLException {
        if (inMemory) {
            if (fallbackProfile != null && fallbackProfile.id() == playerId) {
                int updatedPoints = Math.addExact(fallbackProfile.totalPoints(), points);
                fallbackProfile = new PlayerProfile(playerId, fallbackProfile.name(), updatedPoints, fallbackProfile.language());
                return updatedPoints;
            }
            return points;
        }
        try (Connection connection = connect()) {
            connection.setAutoCommit(false);
            try {
                int currentPoints;
                try (PreparedStatement select = connection.prepareStatement(
                        "SELECT total_points FROM players WHERE id = ?")) {
                    select.setLong(1, playerId);
                    try (ResultSet result = select.executeQuery()) {
                        if (!result.next()) throw new SQLException("The active player record was not found.");
                        currentPoints = result.getInt(1);
                    }
                }
```

```
            int updatedPoints;
            try {
                updatedPoints = Math.addExact(currentPoints, points);
            } catch (ArithmeticException exception) {
                throw new SQLException("The player's score exceeds the
supported range.", exception);
            }
            try (PreparedStatement update = connection.prepareStatement(
                    "UPDATE players SET total_points = ? WHERE id = ?"))
{
                update.setInt(1, updatedPoints);
                update.setLong(2, playerId);
                if (update.executeUpdate() != 1) {
                    throw new SQLException("The active player score could
not be saved.");
                }
            }
            connection.commit();
            return updatedPoints;
        } catch (SQLException | RuntimeException exception) {
            connection.rollback();
            throw exception;
        }
    }
}

private void createSchema() throws SQLException {
```

```
        try (Connection connection = connect();
             Statement statement = connection.createStatement()) {
            statement.executeUpdate("""
                    CREATE TABLE IF NOT EXISTS players (
                        id INTEGER PRIMARY KEY AUTOINCREMENT,
                        display_name TEXT NOT NULL,
                        name_key TEXT NOT NULL UNIQUE,
                        total_points INTEGER NOT NULL DEFAULT 0 CHECK(total_points >= 0),
                        language TEXT NOT NULL,
                        last_played TEXT NOT NULL
                    )
                    """);
        }
    }

    private Connection connect() throws SQLException {
        return DriverManager.getConnection(jdbcUrl);
    }

    private static long findPlayerId(Connection connection, String nameKey) throws SQLException {
        try (PreparedStatement query = connection.prepareStatement(
                "SELECT id FROM players WHERE name_key = ?")) {
            query.setString(1, nameKey);
            try (ResultSet result = query.executeQuery()) {
                return result.next() ? result.getLong(1) : 0;
```

```
            }
        }
    }

    private static Optional<PlayerProfile> findPlayer(Connection
connection, long playerId) throws SQLException {
        try (PreparedStatement query = connection.prepareStatement(
                "SELECT id, display_name, total_points, language FROM
players WHERE id = ?")) {
            query.setLong(1, playerId);
            try (ResultSet result = query.executeQuery()) {
                return result.next() ? Optional.of(readProfile(result)) :
Optional.empty();
            }
        }
    }

    private static PlayerProfile readProfile(ResultSet result) throws
SQLException {
        String languageName = result.getString("language");
        StudyLanguage language;
        try {
            language = StudyLanguage.valueOf(languageName);
        } catch (IllegalArgumentException exception) {
            throw new SQLException("The saved player language is not
supported: " + languageName, exception);
        }
```

```
        return new PlayerProfile(result.getLong("id"),
    result.getString("display_name"),
                result.getInt("total_points"), language);
        }
    }
```

Chapter 11

Conclusion

1. Conclusion

The Cybersecurity Awareness Game developed in this project aims to achieve its objective of creating awareness about cybersecurity and online safety. The system focuses on helping users understand common cyber threats such as phishing, online fraud, password theft, and the misuse of personal information.

The project provides an interactive learning approach to explain important cybersecurity concepts in a simple and understandable manner. It encourages users to recognize online risks, understand the importance of password security, and follow safe internet practices.

One of the key advantages of this project is its educational approach. The game aims to make cybersecurity learning more interesting and accessible, especially for users who have limited knowledge of online threats. It also emphasizes the importance of protecting personal information and behaving responsibly while using digital platforms.

Throughout the development process, different stages such as requirement analysis, system planning, design, implementation, and testing were considered to organize the project effectively. These activities help in developing a structured application focused on cybersecurity awareness.

2. Overall Outcome

The project aims to provide the following outcomes:

- Improved awareness of cybersecurity and online safety
- Better understanding of common cyber threats
- Increased awareness of password security
- Knowledge of safe internet practices
- Greater understanding of personal information protection
- An interactive approach to cybersecurity learning

3. Final Remarks

In conclusion, the Cybersecurity Awareness Game serves as an educational application for understanding cybersecurity concepts in a simple and engaging manner. It encourages users to identify potential online risks and adopt safer digital practices.

The project can be further enhanced by adding more cybersecurity topics, additional game levels, updated questions, score tracking, and other interactive features, depending on future requirements.

Thus, the project provides a foundation for promoting cybersecurity awareness and encouraging responsible internet usage.

Chapter 12

References

During the development of this project, online resources, class notes, and practical lab work were referred to for understanding cybersecurity concepts and preparing the Cybersecurity Awareness Game.

Online Resources

- W3Schools – For understanding basic web development concepts, if applicable to the technologies used.
- TutorialsPoint – For learning programming concepts and software development techniques.
- Cybersecurity awareness resources – For understanding common cyber threats, phishing, password security, and safe internet practices.
- Official cybersecurity websites – For learning about online safety and protecting personal information.

These resources helped in understanding cybersecurity concepts and preparing educational content for the project.

Other Sources

- Class notes provided by the teacher
- Practical lab work
- Discussion and guidance from team members

Tools Used

- Code editor or IDE used for development

- Programming languages and software tools used in the project
- Web browser or runtime environment used for testing

Final Note

The project was developed through practical work, research, and testing. These activities helped in gaining a better understanding of cybersecurity concepts and developing an educational application focused on online safety.